

ARES: Augmented Reality and AI Assistance Technologies for Safety and Efficiency Optimization in Explosive Ordnance Exploration

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ABSTRACT

Unexploded ordnance (UXO) detection remains a critical challenge in past and present conflict zones. Magnetometer surveys are a key method for identifying UXO, but require precise, systematic scanning and expert interpretation. The ARES project leverages Augmented Reality (AR) and Artificial Intelligence (AI) to enhance UXO exploration by improving quality, efficiency, and safety. Using AR glasses, the system guides users through survey areas, assisting with lane alignment, walking speed, and maintaining magnetometer stability. AI-driven processing transforms sparse magnetometer data into dense magnetic maps, while suspected UXO points are directly visualized in the AR display. These features streamline on-site navigation and analysis, offering an intuitive decision-support system for field experts while maintaining reliance on human expertise.

Index terms: Human-centered computing–Mixed / augmented reality, Applied computing–Cartography, Explosive ordnance disposal, EOD, Artificial Intelligence.

1 INTRODUCTION

Unexploded Ordnance (UXO) poses a significant challenge worldwide [1], including in Germany, where remnants of bombs and munitions from the Second World War are still being discovered [1]. Many former military training areas or battlefields remain unusable due to the potential presence of hidden explosive devices [2]. Explosive ordnance disposal (EOD) involves the exploration, identification and clearance of explosive ordnance, including war remnants and mines [3]. In countries affected by war EOD is often essential before any area can be developed, whether for buildings or infrastructure.

Magnetometer surveys are a common EOD method for detecting UXO. A magnetometer operator systematically examines the suspected area, generating magnetic data that is used to create magnetometer maps. Evaluators analyze these maps to identify points of interest (POI), which are subsequently cleared in the field by experts. During the entire EOD process, personnel on site lack direct representation of the measurements, the POI, or the suspected munitions buried in the ground.

Uniform systematic scanning of the test area is crucial for the quality of magnetometer data. The operator must move the



Figure 1: Left: User with AR-enhanced magnetometer; Right: User view with augmented lane-keeping assistance (Guide Low with Arrow), displaying the red parallel target line, actual position (gray line), and arrow indicating the correct movement direction.

magnetometer probe at a consistent speed across lanes while maintaining a stable position. Achieving the necessary measurement quality currently requires extensive experience, as running in a straight line over long distances is nearly impossible for humans [4], especially without reference points. Elaborately set up reference markers in the field can aid in this process.

However, locating suspected points using navigation devices during the clearing phase is often not user-friendly. The ARES project [5] aims to investigate and develop augmented reality (AR) and artificial intelligence (AI) assistance systems to optimize the quality, efficiency, and safety of EOD processes [6]. Our system is designed to assist, not replace, human experts in UXO detection. While it enhances efficiency, supports decision-making and opens up potential for digitised applications, it cannot guarantee the detection of all potential UXOs; therefore, the expertise of trained professionals remains indispensable.

2 SYSTEM SETUP

The ARES system (see Figure 1) consists of a magnetometer (SENSYS SBL10), a Meta Quest 3, a Quest 3 controller mounted to the magnetometer used for 3D tracking of the device and a wireless router and a laptop with an Nvidia GPU used as an edge server. A dedicated AR user interface assists during the EOD tasks.

3 EXPLORATION TASK ASSISTANCE

During exploration, our AR system systematically guides the worker through the survey area in a straight line at a constant speed.

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Four orientation and walking speed assistants are implemented in the demonstrator and are currently evaluated in a user study.

3.1 Lane keeping assistance (LKA)

Our AR system provides several LKA methods to help users walk straight on the lane while keeping the magnetometer steady. Each assistant augments the lane boundaries on the ground, a target line and the deviations from this optimal position. The *Guide Low* method projects a target line and the actual position represented as a line 10 cm above the ground. The *Guide High* method displays the target and actual position line at the magnetometer's height. The *Guide High with Laser* show a directional laser or lance to represent the current position for intuitive orientation. Lastly, the *Guide Low with Arrow* adds a large arrow indicating the correct movement direction to the Guide Low method. In all conditions 3D-deviations from the target line are color coded from green (0-5cm deviation) to red (15 cm deviation) on the assistant lines, arrow or laser.

3.2 Walking speed assistance (WSA)

To help users maintain a steady walking speed while operating the magnetometer, our AR system integrates multiple WSA methods. The AR-view can *display real-time speed* in meters per second. Alternatively, *auditory* or *visually pulsating metronome*-like cues provide a walking rhythm and thus speed guidance. Additionally, a 3D virtual sphere can be displayed moving at the desired speed in the AR-view, allowing users to *follow* it for consistent pacing.

4 RECOVERY TASK ASSISTANCE

Currently available AR glasses lack the necessary resources to run the evaluation algorithms, and none of these devices have a direct connection to the mobile network. Additionally, cellular coverage is limited in many sounding areas. Therefore, we adopted an edge-based approach to enable fast data processing. In this setup, the AR glasses receive measurement data from the magnetometer via Bluetooth, map the data to the local coordinate system of the AR glasses, perform a real-time quality assessment, and subsequently forward this data via Wi-Fi to the edge device in the survey area. After AI processing at the edge, the results are transmitted to the headset via Wi-Fi, for augmentation in the field.

4.1 UXO Data AI Analysis for AR-Headsets

Contrary to the current workflow and similar to research projects e.g. [7], we are using AI instead of analytic methods for gridding the sparse magnetometer data to a dense field. Our gridding AI is based on a U-Net [8], similar to ones that are used for segmenting images. With a second U-Net we determine suspicious points inside the survey area directly from the measurement data (see Figure 2). By using AI instead of the analytic methods, we can remove typical measurement and gridding artefacts. Our AI-based dipole localization should help non-experts to locate the potential UXO. The generated dense magnetic maps can then be visualized directly over the suspected area in the AR glasses. The identified suspected UXO locations are symbolized by markers in the AR display.

4.2 AR Visualisations

Displaying the dense magnetometer map in AR glasses can be challenging, as it obscures ground obstacles. Therefore, our system offers several AR result views. Only important points of interest (POI) can be filtered from the result list and augmented as markers in the field. As users approach a designated marker, detailed information—such as measurement date, GPS coordinates, and estimated depth—is dynamically provided above the corresponding point. In addition, the system can overlay the relevant segment of the magnetometer map with adjustable transparency to improve visibility of the underlying ground and enhance context.

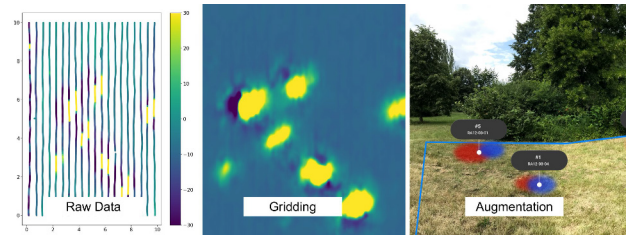


Figure 2: Raw data lanes (left) gridded into a magnetometer map (center) and points of interest (dipole center points) augmented into the field (right).

5 DISCUSSION

Our demonstrator has significant potential for applications in and beyond EOD, extending to various magnetic survey fields. We have developed additional convolutional neural networks (CNNs) for archaeological explorations and adapted the AI and clearance modules for drone-based and pulled surveys. The system is currently under evaluation by expert users.

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